

BREATHE-CC · INTEGRATED EXPOSURE INDEX

IEI additions: where this stands and what I need from you

Decisions from the September 28 meeting, and the calls that still need to be made before the IRB submission

To Dr. Rajesh Melaram

Cc Dr. Donald Warden · Dr. Lei Jin · Dr. Shan-Estelle Brown (Rollins thesis sponsor)

From Aidan Meyers

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With this memo IEI Monthly Follow-up Additions v4 (the full question-by-question spec), the slide deck for Mari, the demo data dictionary, and a clickable preview of the survey

The short version

The additions all sit inside the existing monthly follow-up. After the month the change goes live, a one-home family answers 7 taps and a two-home family 10 to 11, because everything they already told us is shown back to them to confirm. The month it goes live costs 23 taps for one home and 33 to 37 for two.

Everything the group cut on Sunday is out of the dictionary, and everything the group kept is in it. Two things need a decision before I can submit: the medication review (Mari), and how we handle six months of baseline smoking (Driscoll). Both are written out on page 2.

1. What changed after the meeting

Item	Decision	Where it went
Air purifier (C3, C7)	Out of the monthly form	Now a mitigation option on the 3-month safety perception form (P1), with allergen covers, HEPA vacuum, pest control and the rest
Candles and incense (C4, C8)	Cut	Gone
Car windows (C5)	Cut	Gone
Air conditioning running (D2, D4)	Cut	HVAC type at baseline covers it; windows open (D1, D3) stays
Storms and outages (D10)	Kept	Stays monthly. Outage data can't be pulled by address here, so the question is the only route
School or childcare change (D11, D12)	Off the monthly form	Asked once, at the 12-month survey, with the school on file piped in and roughly when it changed
Routine grid columns (A3, A4)	Four columns	Home inside · School or childcare inside · Outside · Part inside, part outside. Sports counts as Outside
Routine grid symbols	Symbol plus word	● Home, ■ School, ○ Outside, ◐ Half and half, so no one reads a key
Smoking	Kept, monthly	Baseline answer piped in, family confirms or updates, same as vaping

Item	Decision	Where it went
Vaping (C13 to C15)	Kept	Indexed on days a week and number of people, which is what the published studies use
Medication review (M1, M2)	Still in, flagged	Decision 1 below
Burden reporting	Taps, not questions	Split into the month it goes live and every month after (Section 2)

2. What this costs a family

Counted by running the dictionary's own branching logic on each case. A tap is one answer; the routine grid counts as seven taps, one per time block.

	One home	Two homes
Every month after the change goes live, nothing changed	7 taps	10 to 11 taps
The month it goes live, or any month something changed	23 taps	33 to 37 taps
The 12-month survey (school question)	8 taps, or 10 if the school changed	Same, plus the two-home items
The most it can ever be	27 taps	48 taps

- **Against today's form:** the monthly survey currently has 38 answerable fields, 19 of them always shown. If the medication review is approved, it removes 3 to 10 medication questions in a month where nothing changed, so an ongoing month lands between -3 and +4 against today.
- **Why the go-live month is the big one:** it is the only month a family enters a routine from scratch. From then on the routine is shown back to them and they confirm it, and a missed month does not reset anything.
- **Families get told first.** A text and email go out with that month's invitation explaining that the survey is longer once and shorter after. Draft wording, English and Spanish, is in Appendix A of the full document.

3. Decisions I need

The first two block the IRB submission. The rest I can carry, but I would rather have your read.

1. The medication review (M1, M2) — Mari's call

What it does: the monthly form shows the medicines the family reported last time and asks "is this still what your child is taking?" Yes skips the existing medication questions for that month; No shows them exactly as they are today. Nothing about the existing questions changes except the branching that hides them when the family says nothing changed.

Dr. Warden's two objections: it changes how an existing section is presented in the middle of the study, which is a harder IRB argument than adding new questions; and medication reporting may look more complete after the switch than before, creating a sub-cohort effect.

Mitigations: medication is EHR-verifiable every month, the switch month is flagged in every record, and I run a before-and-after sensitivity analysis on medication reporting so any jump is visible rather than hidden. Removing it later costs two fields and restoring one line of branching.

What I recommend: Submit it to Mari with the objections and mitigations stated plainly, and drop it without argument if she is uneasy. It is the single biggest burden reduction in the proposal, but it is the only piece that touches an existing section.

2. Six months of baseline smoking — for Driscoll

Baseline asked about smoking once, and for the earliest participants that answer is now six months old. Vaping was never asked. The proposal pipes the baseline answer into the first monthly survey and asks the family to confirm or update it, then repeats that check monthly.

The question for Driscoll: is confirming a six-month-old answer acceptable, or should smoking start fresh at the first monthly survey and be treated as prospective only? Imputing the earlier months is a poor option, because smoking is a count that tends to persist and then stop rather than drift, so imputation would invent a pattern.

What I recommend: Ask Driscoll directly, and flag every back-dated or confirmed-from-baseline value in the data either way, with a sensitivity analysis comparing the months before and after the switch.

3. The duplicated indoor-air questions on the 3-month form

The safety perception forms ask about mold, damp spots and water damage over three months, and the monthly form asks the same three things every month. That is the room we have for the mitigation questions (P1 to P5) at no net cost in question count.

Hiding the three duplicates is a change to existing fields, so it is the team's call, not mine. Deleting them would delete their data, so the proposal uses @HIDDEN if you agree, which keeps everything already collected.

What I recommend: Hide the three duplicates on both safety perception forms and add P1 to P5. That answers the point about mitigation measures without adding to the monthly burden.

4. Dust mites, and traffic near the home

Your dust mite point is partly covered: P1 asks whether the household uses allergen-proof mattress or pillow covers, which is the intervention with the strongest evidence in the 2025 systematic review. What it does not ask is carpet and bedding directly.

Dr. Warden's vehicles-per-hour variable is not in this proposal because we can derive it: the homes and schools are geocoded, so road class and traffic counts can come from TxDOT AADT data rather than from families.

What I recommend: Leave the mite question at the mitigation level in P1 unless you want a direct carpet and bedding item, and derive traffic from TxDOT rather than asking. Both are cheap to add later; neither costs a participant tap now.

5. Two formatting questions I can go either way on

Car time (A5): five bands ("less than 15 minutes" to "more than 2 hours"), or two boxes for hours and minutes that I treat as a share of the day. Bands are faster and less accurate.

The routine grid: one place per time block (current), or check all that apply and split the block evenly between them.

What I recommend: Keep the bands and one place per block for the first version. Both are easy to widen later, and the grid is where the burden is concentrated.

4. The other points raised, and how they are handled

Raised	How it is handled
Dr. Jin: new questions add missing values, and the variable count may outgrow the sample	The items feed two sub-indices, not a model with one term per question, so the analysis sees the index. Most blanks are structural (a one-home family never sees Section B) and are treated as not applicable rather than imputed. Where the count is still too high, items collapse into constructs, for example one indoor-source term from cooking, smoking and vaping. Section 7 of the full document.
Dr. Jin: generative or simulated data for validation	Noted as the route for the methods paper rather than the thesis. The thesis reports the index against the nearest-monitor approach on the cohort we have and states the limitation.
Dr. Melaram: pet breed was never collected	Not proposed as a new question. If it matters for a specific analysis, it is a short manual follow-up with the small number of families who reported a dog or cat, not a monthly item.
Dr. Warden: back-dating school and routine	School calendars are published, so "since the start of this school year" can be dated from the district calendar rather than from the family. Nothing is applied before enrollment, and back-dated values carry a flag.
Carry-forward when a month is missed	A hidden instrument keeps each family's latest answers and the date they were given, so a missed month shows the last saved answers rather than a blank form. No external service is needed; it is calculated inside REDCap, with a nightly API job as a fallback if the cross-event calculations lag.

5. What I am doing next

1. Copy the REDCap project with the records, disable every invitation and alert in the copy, and load the demo dictionary so the piping can be seen with real answers.
2. Test the carry-forward in the copy: first month, a confirm month, a missed month, a season change, and each custody pattern.
3. Finish the slide deck for Mari in the structure Dr. Warden described: the changes, the added taps at go-live versus ongoing, then A, B, C and D with what changed, why, and the value add.
4. Send the printed surveys and the data dictionary for the IRB packet once decisions 1 and 3 are settled.
5. Send the new question text for translation, along with the participant notice in Appendix A.

Timing: I can have the copied project and the deck ready for Thursday. If decisions 1 and 3 land this week, the IRB packet can follow right after, and translation can run in parallel since the question text does not depend on either decision.